

Web Technologies

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Introduction to Web Technologies

- Web technologies refer to the tools, techniques, and standards used for creating, maintaining, and delivering websites and web applications.
- Web technologies are essential for building interactive, responsive, and secure web solutions that meet needs of users and businesses.
- These include programming languages like HTML, CSS, and JavaScript, frameworks and libraries for enhancing functionality and design,
- And also back-end systems and databases that store and manage data, and protocols and services that facilitate communication and data transfer over the internet.

Website and Web Application

- **Website:** A website is a collection of interlinked web pages hosted on a web server, accessed through a browser. Websites primarily focus on providing information to users in a static or semi-static format.

Examples: Personal blogs, news sites, company websites, and online portfolios.

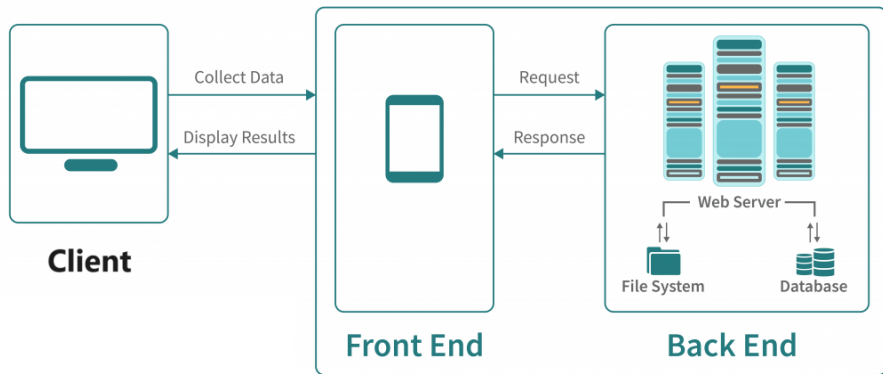
- **Web Application:** A web application is a dynamic, interactive platform that allows users to perform specific tasks online. Web applications provide functionalities similar to desktop software but run within a web browser.

Examples: Online banking systems, social media platforms, e-commerce sites, and web-based productivity tools (like Google Docs).

Website and Web Application

Website	Web Application
Primarily informational	Task-oriented, functionality-focused
Minimal interaction with users	High level of user interaction
It is Static or semi-static	It is Dynamic and frequently updated
Generally simpler in design and functionality	More complex, requiring advanced development
Little on back-end processing	Extensive back-end processing
It is less Customizable and Scalable	It is highly customizable.

Web Application Architecture



Languages and Technologies

Front End Language and Technologies:

- **HTML (HyperText Markup Language):** The foundational language used to structure and format content on the web. HTML defines the structure of a webpage.
- **CSS (Cascading Style Sheets):** A styling language used to define the appearance and layout of HTML elements. CSS allows developers to apply colors, fonts, and other design elements to make websites visually appealing.
- **JavaScript:** A powerful scripting language used to create dynamic and interactive web content. JavaScript enables features such as form validation, animations, and asynchronous data loading without reloading the entire webpage.
- **Frameworks and Libraries:** Frameworks like bootstrap, Tailwind and Tools like React, Angular, and Vue.js, which provide ready-to-use components and functionality to speed up and simplify the development of complex web interfaces.

Languages and Technologies

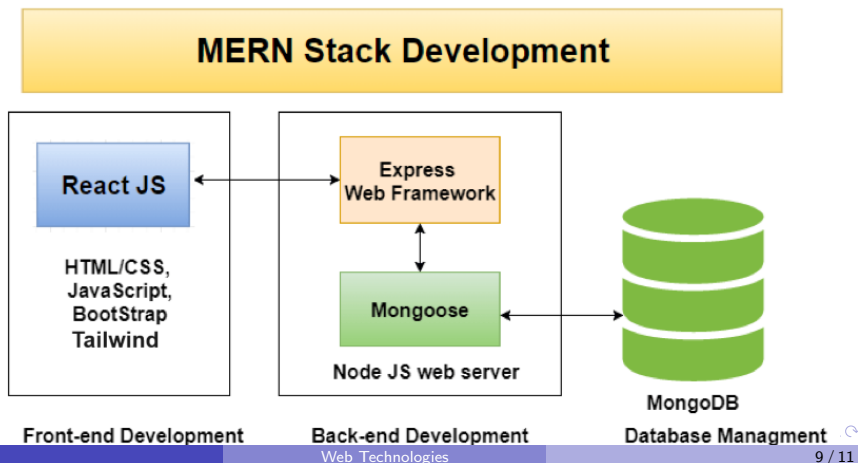
Backend End Language and Technologies:

- **Programming Languages** Languages such as PHP, Python, Ruby, NodeJS and Java are used to develop the logic that runs on the server, handling requests from users, processing data, and generating dynamic content.
- **Databases:** Systems like MySQL, PostgreSQL, MongoDB, and Redis are used to store and manage data in an organized manner, enabling efficient retrieval and manipulation of information.
- **APIs and Web Services:** Application Programming Interfaces (APIs), ExpressJs allow different software applications to communicate and exchange data. RESTful APIs and GraphQL are commonly used to provide data and functionality to web applications.
- **Web Servers:** Software like Apache, Nginx, and Microsoft IIS that handle HTTP requests from clients and deliver the appropriate content.

MERN Stack Architecture

MERN is a pre-built technology stack based on JavaScript technologies. MERN stands for MongoDB, Express, React, and Node, after the four key technologies that make up the stack.

Architecture:



Front End

Front End:

The basic languages of Front-End Development are:

- HTML
- CSS
- JavaScript

The popular frameworks of Front-End Development are:

- Bootstrap
- Tailwind
- React

HTML

HAPPY TO LEARN HTML